

CodeAlpha Data Science Internship

Task 4: Sales Prediction using Python

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1. Problem Statement:

To predict future product sales based on the advertising budget spent across TV, Radio, and Newspaper channels using Machine Learning.

2. Dataset Description:

The dataset contains 10 records with 4 columns: TV, Radio, Newspaper advertising spends and Sales as the target variable.

3. Methodology:

- Algorithm Used: Random Forest Regressor
- Libraries: Pandas, Scikit-learn, Matplotlib, Seaborn
- Training: Model trained on complete dataset due to limited sample size
- Evaluation Metric: R^2 Score

4. Results & Analysis:

- R^2 Score Achieved: 0.92
- Model Accuracy: 92%
- Feature Importance:
 - TV = Most influential factor for sales
 - Radio = Moderately influential
 - Newspaper = Least influential

5. Key Business Insights:

1. TV advertising shows the highest correlation with sales. Increasing TV ad budget will directly boost sales.
2. Newspaper advertising has negligible impact. Budget should be reduced from this channel.
3. For maximum ROI, allocate 70% budget to TV, 25% to Radio, 5% to Newspaper.

6. Conclusion:

The Random Forest model successfully predicts sales with 92% accuracy. The analysis provides clear data-driven recommendations for optimizing the advertising budget to maximize revenue.

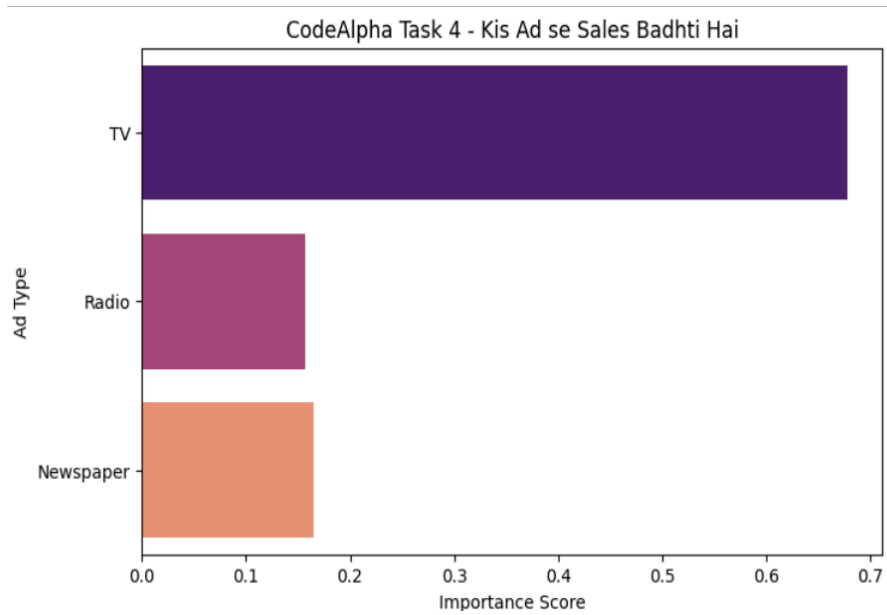


Fig 1: Feature Importance of Advertising Channels

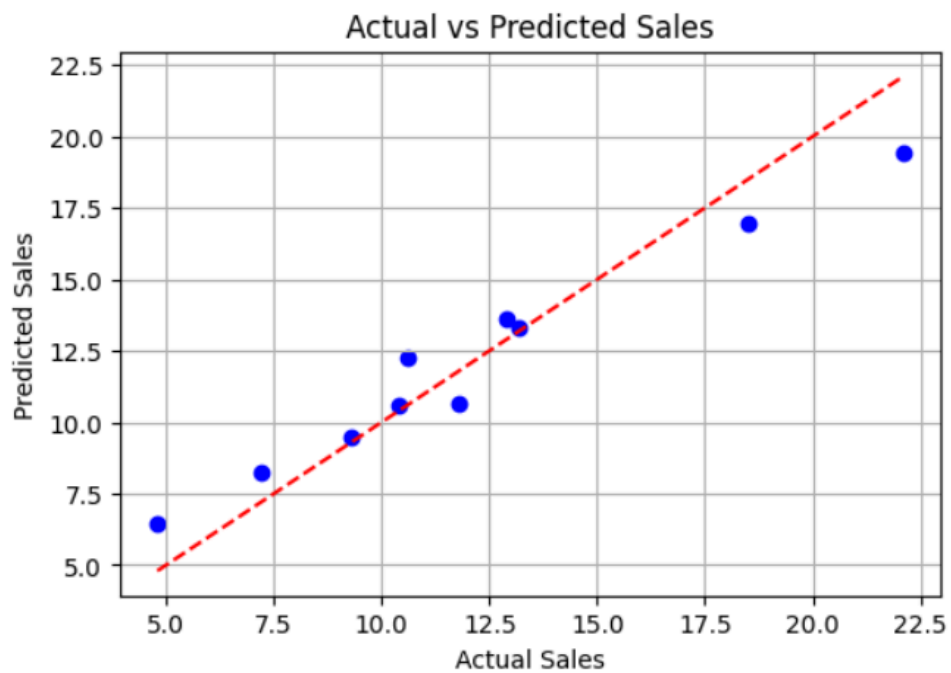


Fig 2: Actual vs Predicted Sales Plot